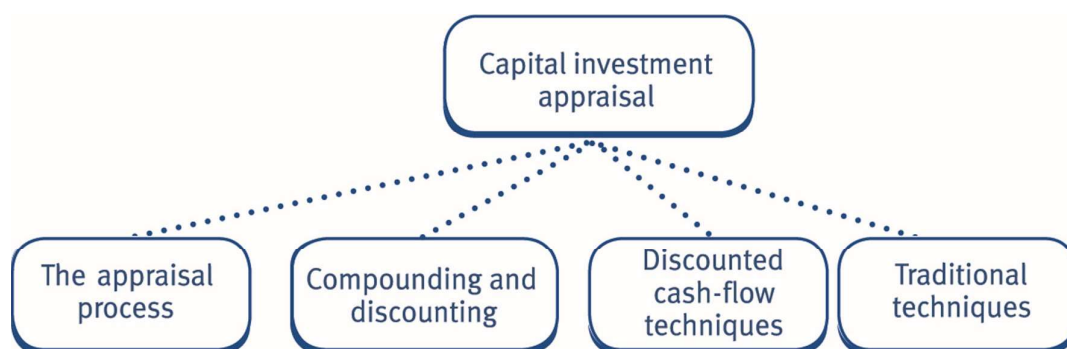




## 1 Capital investment

 When a business spends money on new non-current assets it is known as capital investment or capital expenditure. Spending is normally irregular and for large amounts. It is expected to generate long-term benefits.

Capital investment decisions normally represent the most important decisions that an organisation makes, since they commit a substantial proportion of a firm's resources to actions that are likely to be irreversible.

Many different investment projects exist including:

- replacement of assets
- cost-reduction schemes
- new product/service developments
- product/service expansions
- statutory, environmental and welfare proposals.

## 2 Capital and revenue expenditure

The distinction between capital expenditure and revenue expenditure is important.

### Capital expenditure

Capital expenditure is expenditure incurred in:

- (a) the acquisition of non-current assets required for use in the business and not for resale
- (b) the alteration or significant improvement of non-current assets for the purpose of increasing their revenue-earning capacity.

Capital expenditure is initially shown in the **statement of financial position** as non-current assets. It is then **charged to the statement of profit or loss** over a number of periods, via the **depreciation** charge.

## Capital budgeting

### Revenue expenditure

Revenue expenditure is expenditure incurred in:

- (a) the purchase of assets acquired for conversion into cash (e.g. goods for resale)
- (b) the manufacturing, selling and distribution of goods and the day-to-day administration of the business
- (c) the maintenance of the revenue-earning capacity of the non-current assets (i.e. repairs, etc).

Revenue expenditure is generally charged to the **statement of profit or loss** for the period in which the expenditure was incurred.

In practice, there can be some difficulty in clearly distinguishing between alteration/improvement of non-current assets (capital) and their maintenance (revenue). For example, is the installation of a modern heating system to replace an old inefficient system an improvement or maintenance? However, you should not need to make such decisions in your exam.

### 3 Capital budgeting and investment appraisal



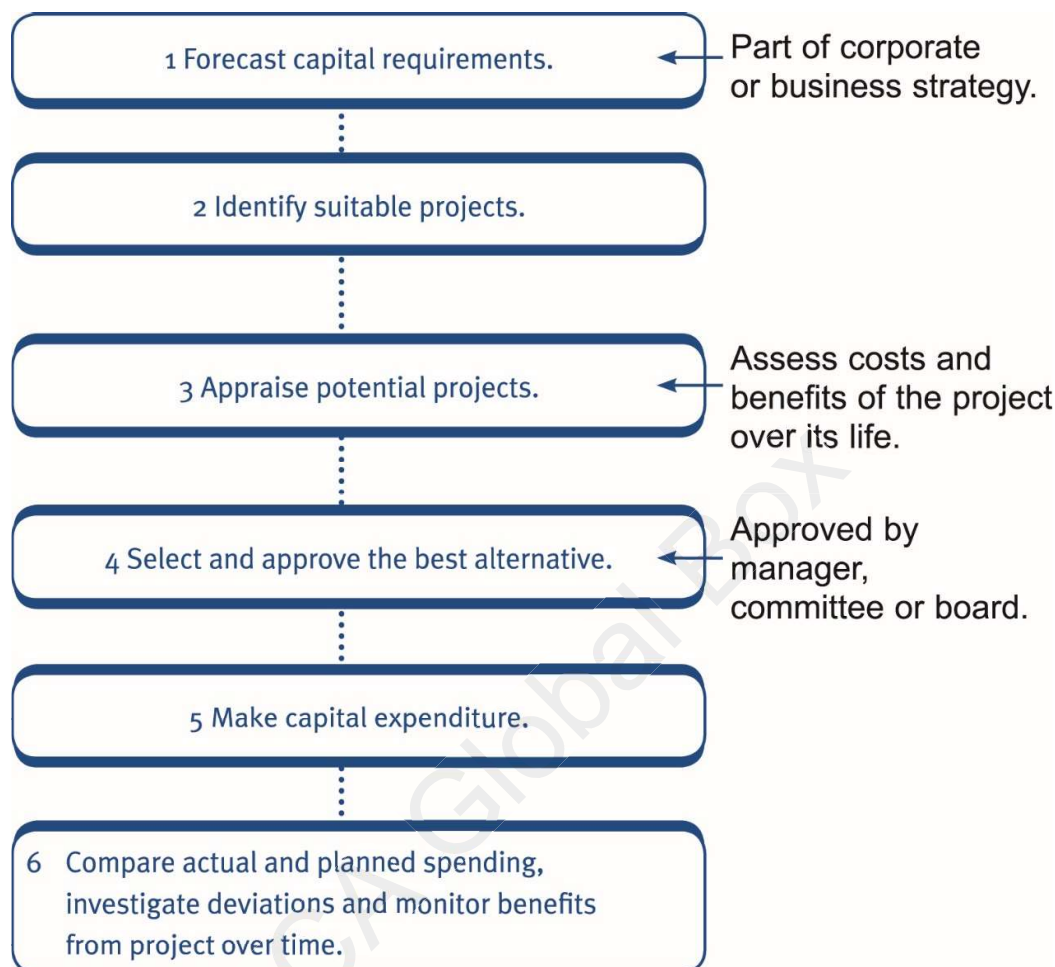
A capital budget:

- is a programme of capital expenditure covering several years
- includes authorised future projects and projects currently under consideration.

One stage in the capital budgeting process is **investment appraisal**. This appraisal has the following features:

- estimates of future costs and benefits over the project's life
- assessment of the level of expected returns earned.

The capital budgeting process consists of a number of stages:



#### 4 Cash flows used for investment appraisal

In capital investment appraisal it is more appropriate to evaluate future **cash flows rather than accounting profits**. Cash and profit are very different. Profit is calculated on the statement of profit or loss and cash is a current asset on the statement of financial position. The differences arise because:

- Revenue is recognised in the statement of profit or loss when it is earned but this is not necessarily when the cash is received.
- Costs are recognised in the statement of profit or loss when they are incurred but this is not necessarily when the cash is paid.
- Non-cash expenses – the statement of profit or loss of a business is charged with a number of non-cash expenses such as depreciation and provisions for doubtful debts. Although these are correctly charged as expenses in the statement of profit or loss, they are not cash flows and will not reduce the cash balance of the business.

## Capital budgeting

- Purchase of non-current assets – these are often large cash outflows of a business but the only amount that is charged to the statement of profit or loss is the annual depreciation charge not the entire cost of the non-current asset.
- Sale of non-current assets – when a non-current asset is sold this will result in an inflow of cash to the business but the figure to appear in the statement of profit or loss is not the cash proceeds but any profit or loss on the sale.
- Financing transactions – some transactions, such as issuing additional share capital and taking out or repaying a loan, will result in large cash flows in or out of the business with no effect on the profit figure at all.

When appraising a possible capital investment it is necessary to use the actual cash flows in and out of the business rather than profits as profits are subjective and cannot be spent.

Cash flows that are appraised should be relevant to or change as a direct result of making a decision to invest. **Relevant cash flows** are:

- future costs and revenues – it is not possible to change what has happened so any relevant costs or revenues are future ones
- cash flows – actual cash coming in or leaving the business not including any non-cash items such as depreciation and notional costs
- incremental costs and revenues – the change in costs or revenues that occurs as a direct result of a decision to invest.



### Relevant cost terms

#### Relevant cost terminology

- **Differential costs** are the differences in total costs or revenues between two alternatives.
- **Opportunity cost** is an important concept in decision making. It represents the best alternative that is foregone in taking the decision. The opportunity cost emphasises that decision making is concerned with alternatives and that the cost of taking one decision is the profit or contribution foregone by not taking the next best alternative.
- **Avoidable costs** are the specific costs associated with an activity that would be avoided if that activity did not exist.

### Non-relevant cost terminology

- **Sunk costs** are past or historical costs which are not directly relevant in decision making, for example, development costs or market research costs.
- **Committed costs** are future costs that cannot be avoided, whatever decision is taken.
- **Non-cash flow costs** are costs which do not involve the flow of cash, for example, depreciation and notional costs. A notional cost is a cost that will not result in an outflow of cash either now or in the future, for example, sometimes the head office of an organisation may charge a 'notional' rent to its branches. This cost will appear in the accounts of the organisation but will not result in a 'real' cash expenditure.
- **General fixed overheads** are usually not relevant to a decision. However, some fixed overheads may be relevant to a decision; for example, stepped fixed costs may be relevant if fixed costs increase as a direct result of a decision being taken.
- **Carry amount of non-current assets** are not relevant costs because like depreciation, they are determined by accounting conventions rather than by future cash flows.



### Illustration 1 – Relevant costs

A company is evaluating a proposed expenditure on an item of equipment that would cost \$160,000.

A technical feasibility study has been carried out by consultants, at a cost of \$15,000, into benefits from investing in the equipment.

It has been estimated that the equipment would have a life of four years, and annual profits would be \$8,000, after deducting annual depreciation of \$40,000 and an annual charge of \$25,000 for a share of the existing fixed cost of the company.

#### Required:

What are the relevant cash flows for this investment?

#### Solution:

The \$160,000 to be spent on the new item of equipment is relevant as it is a future cash flow incurred as a direct result of making the decision.

The \$15,000 already spent on the feasibility study is a sunk cost – it has already been spent – therefore it is not relevant.

Depreciation and apportioned fixed overheads are not relevant. Depreciation is not a cash flow and apportioned fixed overheads represent costs that will be incurred anyway.

## Capital budgeting



### The cost of capital

In the above discussions we referred to the rate of interest. There are a number of alternative terms used to refer to the rate a firm should use to take account of the time value of money:

- cost of capital
- discount rate
- required return.



Whatever term is used, the rate of interest used for discounting should aim to reflect the cost of the finance that will be tied up in the investment.

## 8 Capital investment appraisal

### Appraisal methods

There are three widely used appraisal methods:

- 1 The payback period (using both discounted and non-discounted cash flows).
- 2 Net present value (NPV).
- 3 Internal rate of return (IRR).

All three methods consider the time value of money, assuming the discounted payback method is used. They are known as **discounted cash flow (DCF) techniques**.

## 9 The payback period



The payback period is the time a project will take to pay back the money spent on it. It is based on expected cash flows and provides a measure of liquidity.

It is the time which elapses until the invested capital is recovered. It considers cash flows only. It can be assumed that, with this technique, the cash flows can occur evenly during the year.

### Decision criteria

- Compare the payback period to the company's maximum return time allowed and if the payback is quicker the project should be accepted.
- Faced with mutually-exclusive projects choose the project with the shortest payback.

### Calculation – Constant annual flows

$$\text{Payback period} = \frac{\text{Initial investment}}{\text{Annual cash inflow}}$$

A payback period may not be for an exact number of years. To calculate the payback in years and months you should multiply the decimal fraction of a year by 12 to get the number of months.



### Test your understanding 7

An expenditure of \$2 million is expected to generate net cash inflows of \$500,000 each year for the next seven years.

#### Required

Calculate the payback period for the project?



### Test your understanding 8

A project will involve spending \$1.8 million now. Annual cash flows from the project would be \$350,000.

#### Required

Calculate the payback period for the project.

## Calculations – Uneven annual flows

If cash flows are uneven (a more likely state of affairs), the payback has to be calculated by working out the cumulative cash flow over the life of a project.



### Illustration 6 – Payback period

Minnie Ltd is considering two mutually-exclusive projects with the following details:

#### Project A

|                           |           |     |     |     |     |
|---------------------------|-----------|-----|-----|-----|-----|
| Initial investment        | \$450,000 |     |     |     |     |
| Scrap value in year 5     | \$20,000  |     |     |     |     |
| Year:                     | 1         | 2   | 3   | 4   | 5   |
| Annual cash flows (\$000) | 200       | 150 | 100 | 100 | 100 |

#### Project B

|                           |           |    |    |    |    |
|---------------------------|-----------|----|----|----|----|
| Initial investment        | \$100,000 |    |    |    |    |
| Scrap value in year 5     | \$10,000  |    |    |    |    |
| Year:                     | 1         | 2  | 3  | 4  | 5  |
| Annual cash flows (\$000) | 50        | 40 | 30 | 20 | 20 |

Assume that the initial investment is at the start of the project and the annual cash flows accrue evenly over the year.

#### Required:

Calculate which project the company should select if the objective is to minimise the payback period?

## Capital budgeting

**Solution****Project A**

|        | <b>Cash flow</b> | <b>Cumulative cash flow</b> |
|--------|------------------|-----------------------------|
|        | \$000            | \$000                       |
| Year 0 | (450)            | (450)                       |
| Year 1 | 200              | (250)                       |
| Year 2 | 150              | (100)                       |
| Year 3 | 100              | 0                           |

Payback period = 3 years

**Project B**

|        | <b>Cash flow</b> | <b>Cumulative cash flow</b> |
|--------|------------------|-----------------------------|
|        | \$000            | \$000                       |
| Year 0 | (100)            | (100)                       |
| Year 1 | 50               | (50)                        |
| Year 2 | 40               | (10)                        |
| Year 3 | 30               | 20                          |

Project B requires \$10,000 during year 3 to payback. Over the 3rd year \$30,000 cash is being received. Assuming that cash accrues evenly over the year it will take 1/3 of a year to recoup the remaining cash to payback.  $1/3 \times 12 \text{ months} = 4 \text{ months}$ .

Payback period = 2 years 4 months

**Project B should be accepted**

**Discounted payback**

One of the major criticisms of using the payback period is that it does not take into account the time value of money. The discounted payback technique attempts to overcome this criticism. The technique is identical – but the present value of the cash flow is calculated before calculating the cumulative cash flow.

**Illustration 7 – Discounted payback period**

Minnie Ltd is considering two mutually-exclusive projects with the following details:

**Project A**

|                           |           |     |     |     |     |
|---------------------------|-----------|-----|-----|-----|-----|
| Initial investment        | \$450,000 |     |     |     |     |
| Scrap value in year 5     | \$20,000  |     |     |     |     |
| Year:                     | 1         | 2   | 3   | 4   | 5   |
| Annual cash flows (\$000) | 200       | 150 | 100 | 100 | 100 |



**Project B**

Initial investment \$100,000

Scrap value in year 5 \$10,000

|                           |    |    |    |    |    |
|---------------------------|----|----|----|----|----|
| Year:                     | 1  | 2  | 3  | 4  | 5  |
| Annual cash flows (\$000) | 50 | 40 | 30 | 20 | 20 |

Assume that the initial investment is at the start of the project and the annual cash flows accrue evenly over the year.

**Required:**

Calculate the discounted payback period for both projects if the relevant cost of capital is 10%.

**Solution**

| Year | Discount factor<br>10% | Project A          |                        |                               | Project B          |                        |                               |
|------|------------------------|--------------------|------------------------|-------------------------------|--------------------|------------------------|-------------------------------|
|      |                        | Cash flow<br>\$000 | Present value<br>\$000 | Cumulative cash flow<br>\$000 | Cash flow<br>\$000 | Present value<br>\$000 | Cumulative cash flow<br>\$000 |
| 0    | 1.000                  | (450)              | (450)                  | (450)                         | (100)              | (100)                  | (100)                         |
| 1    | 0.909                  | 200                | 181.8                  | (268.2)                       | 50                 | 45.45                  | (54.55)                       |
| 2    | 0.826                  | 150                | 123.9                  | (144.3)                       | 40                 | 33.04                  | (21.51)                       |
| 3    | 0.751                  | 100                | 75.1                   | (69.2)                        | 30                 | 22.53                  | 1.02                          |
| 4    | 0.683                  | 100                | 68.3                   | (0.9)                         |                    |                        |                               |
| 5    | 0.621                  | 120                | 74.52                  | 73.62                         |                    |                        |                               |

Project A now pays back in just over 4 years and Project B in just under 3 years. Project B is still preferable to Project A but the payback period has increased as time value of money is applied to the cash flows.

| Advantages                                                                                                                                                                                                                                                                                                                     | Disadvantages                                                                                                                                                                                                                                                                                       |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>Simple to understand</li> <li>Payback is a simple measure of risk. Firms selecting projects on the basis of payback periods may avoid liquidity problems</li> <li>Uses cash flows, not subjective accounting profits</li> <li>Emphasises the cash flows in the earlier years</li> </ul> | <ul style="list-style-type: none"> <li>Is not a measure of absolute profitability</li> <li>Ignores the time value of money. <b>Note:</b> A discounted payback period may be calculated to overcome this problem</li> <li>Does not take into account cash flows beyond the payback period</li> </ul> |



### Advantages and disadvantages of Payback

Payback has a number of advantages

- As a concept payback is easily understood and is easily calculated.
- Rapidly changing technology. If new plant is likely to be scrapped in a short period because of obsolescence, a quick payback is essential.
- Improving investment conditions. When investment conditions are expected to improve in the near future, attention is directed to those projects that will release funds soonest, to take advantage of the improving climate.
- Payback favours projects with a quick return. It is often argued that these are to be preferred for three reasons.
  - Rapid project payback leads to rapid company growth – but in fact such a policy will lead to many profitable investment opportunities being overlooked because their payback period does not happen to be particularly swift.
  - Rapid payback minimises risk – the logic being that the shorter the payback period, the less there is that can go wrong. Not all risks are related to time, but payback is able to provide a useful means of assessing time risk. It is likely that earlier cash flows can be estimated with greater certainty.
  - Rapid payback maximises liquidity – but liquidity problems are best dealt with separately, through cash forecasting.
- Payback uses cash flows, rather than profits, and so is less likely to produce an unduly optimistic figure distorted by assorted accounting conventions which might permit certain costs to be carried forward and not affect profit initially.

The disadvantages of payback are:

- Project returns may be ignored. In particular, cash flows arising after the payback period are totally ignored.
- Timing of cash flows are ignored. Cash flows are effectively categorised as pre-payback or post-payback, but no more accurate measure is made.
- The time value of money is ignored (unless discounted payback is used).
- There is no objective measure as to what length of time should be set as the minimum or maximum payback period. Investment decisions are therefore subjective.
- Payback takes no account of the effects on business profits and periodic performance of the project, as evidenced in the financial statements. This is critical if the business is to be reasonably viewed by users of the accounts.

## 10 Net present value (NPV)



The NPV of an investment represents the net benefit or loss of benefit in present value terms for an investment opportunity.



A positive NPV represents the surplus funds earned on a project. This means that it tells us the impact on shareholder wealth.

### Decision criteria

- Any project with a positive NPV is viable.
- Projects with a negative NPV are not viable.
- Faced with mutually-exclusive projects, choose the project with the highest NPV.



### What does the NPV actually mean

Suppose, in an investment problem, we calculate the NPV of certain cash flows at 12% to be – \$97, and at 10% to be zero, and yet at 8% the NPV of the same cash flows is + \$108. Another way of expressing this is as follows.

- If the funds were borrowed at 12% the investor would be \$97 out of pocket – i.e. the investment earns a yield below the cost of capital.
- If funds were borrowed at 10% the investor would break even – i.e. the investment yields a return equal to the cost of capital.
- If funds were borrowed at 8% the investor would be \$108 in pocket – i.e. the investment earns a return in excess of the cost of capital.

In other words, a positive NPV is an indication of the surplus funds available to the investor now as a result of accepting the project.



### Illustration 8 – Net present value (NPV)

Minnie Ltd is considering two mutually-exclusive projects with the following details:

#### Project A

|                           |           |     |     |     |     |
|---------------------------|-----------|-----|-----|-----|-----|
| Initial investment        | \$450,000 |     |     |     |     |
| Scrap value in year 5     | \$20,000  |     |     |     |     |
| Year:                     | 1         | 2   | 3   | 4   | 5   |
| Annual cash flows (\$000) | 200       | 150 | 100 | 100 | 100 |

## Capital budgeting

**Project B**

Initial investment \$100,000

Scrap value in year 5 \$10,000

|                           |    |    |    |    |    |
|---------------------------|----|----|----|----|----|
| Year:                     | 1  | 2  | 3  | 4  | 5  |
| Annual cash flows (\$000) | 50 | 40 | 30 | 20 | 20 |

Assume that the initial investment is at the start of the project and the annual cash flows accrue evenly over the year.

**Required:**

Calculate the Net Present Value to the nearest \$000 for Projects A and B if the relevant cost of capital is 10%.

**Solution**

|      |                 | Project A       |                     | Project B       |                     |
|------|-----------------|-----------------|---------------------|-----------------|---------------------|
| Year | Discount factor | Cash flow \$000 | Present value \$000 | Cash flow \$000 | Present value \$000 |
| 0    |                 | (450)           | (450)               | (100)           | (100)               |
| 1    | 0.909           | 200             | 182                 | 50              | 45                  |
| 2    | 0.826           | 150             | 124                 | 40              | 33                  |
| 3    | 0.751           | 100             | 75                  | 30              | 23                  |
| 4    | 0.683           | 100             | 68                  | 20              | 14                  |
| 5    | 0.621           | 120             | 75                  | 30              | 19                  |
|      |                 | NPV =           | 74                  | NPV =           | 34                  |

**Test your understanding 9**

An organisation is considering a capital investment in new equipment. The estimated cash flows are as follows.

| Year | Cash flow \$ |
|------|--------------|
| 0    | (240,000)    |
| 1    | 80,000       |
| 2    | 120,000      |
| 3    | 70,000       |
| 4    | 40,000       |
| 5    | 20,000       |

The company's cost of capital is 9%.

**Required**

Calculate the NPV of the project to assess whether it should be undertaken.

| Advantages                                                                                                                                                                                                                                                                                                                              | Disadvantages                                                                                                                                                                         |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>Does consider the time value of money</li> <li>It is a measure of absolute profitability</li> <li>Considers cash flows</li> <li>It considers the whole life of the project</li> <li>A company selecting projects on the basis of NPV maximisation should maximise shareholders wealth</li> </ul> | <ul style="list-style-type: none"> <li>Fairly complex</li> <li>Not well understood by non-financial managers</li> <li>It may be difficult to determine the cost of capital</li> </ul> |



### Advantages and disadvantages of NPV

When appraising projects or investments, NPV is considered to be superior to other methods. This is because it:

- considers the time value of money – discounting cash flows to their present value takes account of the impact of interest, inflation and risk over time.
- is an absolute measure of return – the NPV of an investment represents the potential surplus raised by the project. This allows a business to plan more effectively.
- is based on cash flows not profits – the subjectivity of profits makes them less reliable than cash flows and therefore less appropriate for decision making.
- considers the whole life of the project – NPV takes account of all relevant flows associated with the project. Discounting the flows takes account of the fact that later flows are less reliable.
- should lead to maximisation of shareholder wealth. If the cost of capital reflects the investors' (i.e. shareholders') required return, then the NPV reflects the theoretical increase in their wealth. For a company, this is considered to be the primary objective of the business.

However, there are some potential drawbacks:

- It is difficult to explain to managers. To understand the meaning of the NPV calculated requires an understanding of discounting. The method is not as intuitive as techniques such as payback.
- It requires knowledge of the cost of capital. The calculation of the cost of capital is, in practice, more complex than identifying interest rates. It involves gathering data and making a number of calculations based on that data and some estimates. The process may be deemed too protracted for the appraisal to be carried out.
- It is relatively complex. For the reasons explained above, NPV may be rejected in favour of simpler techniques.

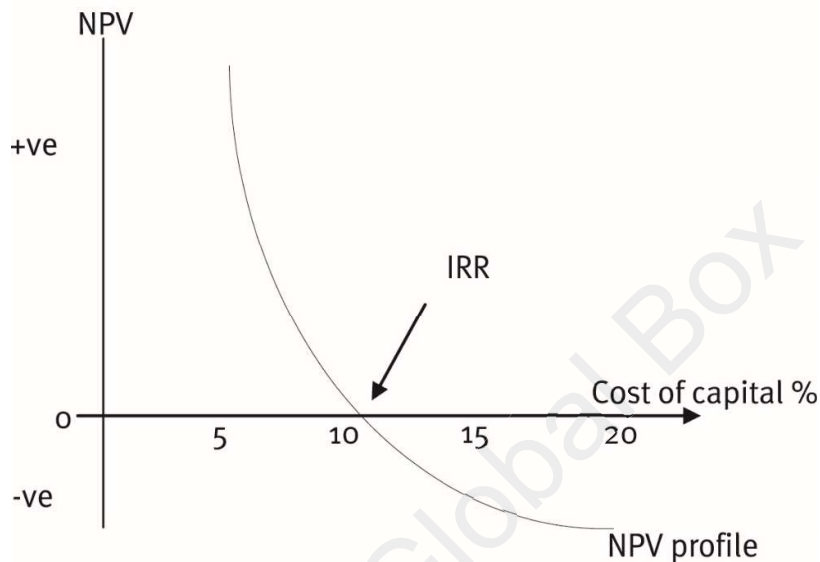
## Capital budgeting

## 11 Internal rate of return (IRR)



This is the rate of return, or discount rate, at which a project has a NPV of zero.

Graph showing the NPV of one project appraised at different discount rates

**Decision criteria**

- If the IRR is greater than the company's cost of capital the project should be accepted.
- Faced with mutually-exclusive projects choose the project with the higher IRR.

**Calculating the IRR (using linear interpolation)**

The steps in linear interpolation are:

- Calculate two NPVs for the project at two different costs of capital
- Use the following formula to find the IRR:

**FORMULA FOR IRR**

$$\text{IRR} = L + \frac{N_L}{N_L - N_H} \times (H - L)$$

**LEARN**

where:

L = Lower rate of interest

H = Higher rate of interest

$N_L$  = NPV at lower rate of interest

N = NPV at higher rate of interest.



### Illustration 9 – Internal rate of return (IRR)

Net cash flows from Project A are:

| Year | Net cash flow \$000 |
|------|---------------------|
| 0    | (450)               |
| 1    | 200                 |
| 2    | 150                 |
| 3    | 100                 |
| 4    | 100                 |
| 5    | 120                 |

The NPV of Project A at a discount rate of 10% is \$73,620.

#### Required:

Calculate the internal rate of return of Project A.

#### Solution

| Year | Discount factors at 20% | Project A           |          |
|------|-------------------------|---------------------|----------|
|      |                         | Net cash flow \$000 | PV \$000 |
| 0    | 1.000                   | (450)               | (450)    |
| 1    | 0.833                   | 200                 | 167      |
| 2    | 0.694                   | 150                 | 104      |
| 3    | 0.579                   | 100                 | 58       |
| 4    | 0.482                   | 100                 | 48       |
| 5    | 0.402                   | 120                 | 48       |
|      |                         | NPV @ 20% =         | (25)     |

$$IRR = L + \frac{N_L}{N_L - N_H} \times (H - L)$$

$$IRR = 10 + \frac{74}{74 - -25} \times (20 - 10)$$

$$IRR = 10 + \frac{74}{99} \times (10)$$

$$IRR = 10 + (0.747 \times 10)$$

$$IRR = 17.5\%$$

## Capital budgeting

**Further explanation of IRR**

Using the NPV method, PVs are calculated by discounting cash flows at a given cost of capital, and the difference between the PV of costs and the PV of benefits is the NPV. In contrast, the IRR method of DCF analysis is to calculate the exact DCF rate of return that the project is expected to achieve.

If an investment has a positive NPV, it means it is earning more than the cost of capital. If the NPV is negative, it is earning less than the cost of capital. This means that if the NPV is zero, it will be earning exactly the cost of capital.

Conversely, the percentage return on the investment must be the rate of discount or cost of capital at which the NPV equals zero. This rate of return is called the IRR or the DCF yield and if it is higher than the target rate of return then the project is financially worth undertaking.

**Estimating the IRR**

An investment has a NPV with a discount rate of 10% of \$4,400 and a NPV at a discount rate of 20% of –\$31,000.

The IRR will be closer to 10% than 20% as \$4,400 is closer to zero than – \$31,000.

We can estimate the IRR:

- There is a difference of 10% between the 10% and 20% used for the NPV calculations
- There is a difference in NPV value of  $\$31,000 + \$4,400 = \$35,400$
- Each percentage change is approximately  $(31,000 + 4,400)/10 = \$3,540$
- For the project to break even \$4,400 is required, this is represented by  $\$4,400/\$3,540 = 1.24\%$
- The IRR can be estimated as  $10\% + 1.24\% = 11.24\%$ .



For examination purposes, the choice of rates to estimate the IRR is less important than your ability to perform the calculation to estimate it.

**Test your understanding 10**

You are given the following:

At 20% the NPV is \$8,510

At 30% the NPV is – \$9,150

**Required:**

Calculate the internal rate of return.



**Test your understanding 11**

A potential project's predicted cash flows gives a NPV of \$50,000 at a discount rate of 10% and –\$10,000 at a rate of 15%.

**Required**

Calculate the IRR.

**Test your understanding 12**

Identify the correct explanation of the internal rate of return – it is the interest rate that equates the present value of expected future net cash flows to:

- A the initial cost of the investment outlay
- B the depreciation value of the investment
- C the terminal (compounded) value of future cash receipts
- D the firm's cost of capital

| Advantages                                                                                                                                                                                                                                                                                                                                                                                                                                                | Disadvantages                                                                                                                                                                                                                                                                                                                                                    |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>Does consider the time value of money</li> <li>It is a percentage so should be easily understood by nonfinancial managers</li> <li>Considers cash flows</li> <li>It considers the whole life of the project</li> <li>It can be calculated without reference to the cost of capital</li> <li>A company selecting projects where the IRR exceeds the cost of capital should increase shareholders' wealth</li> </ul> | <ul style="list-style-type: none"> <li>It is not a measure of absolute profitability</li> <li>Interpolation only provides an estimate of the true IRR</li> <li>Fairly complicated to calculate – although spreadsheets now have built-in programs</li> <li>The IRR of projects may conflict with the NPV. If this occurs the NPV must take precedence</li> </ul> |